2025 28th International Conference on Computer and Information Technology (ICCIT)
19-21 December 2025, Cox's Bazar, Bangladesh

Explainable AI-Driven Ensemble Learning Framework for PCOS Diagnosis Using AIM-PDCF and QuantumGraphRFE Feature Selection

Md. Zubayer Ahmad Shibly
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
zubayer0ahmad@gmail.com

Inamul Haque Inam
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
inamulhaqueinam5@gmail.com

Ashfaqul Islam
Dept. of Computer Science
Baylor University
Texas, USA
tonmoy.icche@gmail.com

Abdul Kadar Muhammad Masum
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
akmmasum@yahoo.com

Khandaker Mohammad Mohi Uddin*
Dept. of Computer Science and Engineering
Southeast University
Dhaka, Bangladesh
jilanicsejnu@gmail.com

2025 28th International Conference on Computer and Information Technology (ICCIT) | 979-8-3315-7867-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCIT68739.2025.11491104

***Abstract*— Polycystic Ovary Syndrome (PCOS) is a common, multifaceted endocrine disorder in women of reproductive age where early diagnosis has a significant impact on long-term health consequences. This paper's novel methodology uses a stacking ensemble model for PCOS diagnosis, distinguished by two primary innovations: a use of the Advanced Intelligent Medical Pattern Discovery and Clustering Framework (AIM-PDCF) as data preprocessing tool and a hybrid QuantumGraphRFE model for quantum inspired feature selection. The models were trained and tested extensively with a nested cross-validation strategy to ensure its robustness. KNN-XGB ensemble reached state-of-the-art for the benchmark dataset, with 96.58% accuracy, 96.60% F1-Score, and particularly outstanding 99.59% ROC-AUC. A key element is clinical interpretability; by employing Explainable AI (XAI) techniques including LIME (Global Interpretability Plot, Local Explanation Plot) and SHAP (Waterfall Plot, Heatmap), we substantiate that predictions are influenced by clinically relevant biomarkers such as follicle count, menstrual irregularity, and weight gain. Our proposed model excels existing models in diagnostic accuracy, making it a transparent and reliable tool to assist clinicians in the early detection of PCOS.**

Keywords: Polycystic Ovary Syndrome (PCOS), Stacking Ensemble, Quantum Approximate Quality Assessment, Explainable AI, Clinical Decision Support, Predictive Diagnostics.

I. INTRODUCTION

Polycystic ovary syndrome (PCOS) is a frequent endocrine disorder that affects women in the reproductive period and is manifested by anabolic hormonal disorders, irregular menstruation (oligomenorrhea or amenorrhea), and many cysts in the ovaries [1]. This condition is a common source of chronic anovulation, the cause of infertility, and may be a long-term contributor to insulin resistance, type II diabetes, risk of cardiovascular disease, and psychological complications. The problem of PCOS is of great sociomedical importance as of 2025, there are more than 100 million patients around the world [2]. In addition, challenges of PCOS result in high levels of mortality; hence, cardiovascular diseases, a complication of PCOS, are the leading causes of death among women, with one-third of the female deaths being attributed to the condition in the world [3]. The early diagnosis and dynamic treatment of PCOS has high clinical importance. Recent developments in machine learning (ML) and artificial intelligence (AI) have transformed predictive accuracy to a level where one can predict PCOS at a prodromal stage. These computational models examine a variety of clinical, biochemical and imaging biomarkers to identify minor pathophysiological aberrations before the onset of symptoms. It is a synergistic combination of heterogeneous algorithms and horizontal stack hits which reinforces diagnostic moisture and reliability [4-5]. Simultaneously, Exceptional feature selection heuristics are used to extract the most important biomarkers and remove noise, maximizing the discriminative power. Application of the ensemble learning and feature selection combination theretofore provides predictive systems not only identifying the PCOS phenotype difference among standard populations but also showing causal determinations [6]. Artificial intelligence in diagnostics provides clinicians with high-resolution risk profiling that can be used to implement proactive, individual-patient interventions leading to a new era of precision gynecology.

- In this research, we developed AIM-PDCF's novel categorical converter, a unique medical-aware system that automatically transforms continuous features into optimal categorical representations using distribution analysis and adaptive binning strategies.
- This study implemented the complete AIM-PDCF pipeline, an advanced framework combining attention autoencoder + UMAP + ensemble clustering for automated medical pattern discovery and comprehensive evaluation.
- The proposed research involves developing a QuantumGraph-RFE model that fuses quantum-inspired

979-8-3315-7867-1/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:55:55 UTC from IEEE Xplore. Restrictions apply.

feature scoring, MST-based relationship preservation, and RFECV through iterative optimization with quantum tunneling for enhanced multi-paradigm feature selection with $O(d^2 \times \text{max_iterations} + d^2 \log d + \text{RFECV_complexity})$ complexity.

- The proposed research involves constructing six heterogeneous ensemble models (RF–ETC, ETC–XGB, RF–KNN, KNN–ETC, RF–XGB, KNN–XGB) using Logistic Regression (LR) as the meta-learner, with model evaluation through nested cross-validation, systematic hyperparameter tuning (7 parameters × 3 options), and interpretability ensured via XAI.

II. Literature Review

During the last several years, ML techniques and metaheuristic approaches to predicting and diagnosing PCOS were investigated. These studies indicate the potential of analytical software in the improvement of diagnostic accuracy but they also underline the shortcomings of methodological rigor and interpretation. Sreejith et al. [7] suggested the application of a clinical support framework based on the Red Deer Algorithm (RDA) as a feature selecting process embedded in a Random Forest (RF) wrapper. Out of 37 attributes, 20 were retained, providing 89.81 percent accuracy, 89.73 percent sensitivity and 90.43 percent specificity when used under 10-fold cross-validation. The study did not integrate nested validation or XAI, however, and examined only predictive performance. Rahman et al. [8] provided a web-interface that uses a Mutual Information (MI) model to rank features and tune hyperparameters and utilized 12 out of 45 features. Although only one 80-20 hold out split was utilised, the model still performed well with RF returning a maximum AUC of 0.9075 and AdaBoost showing 94 percent accuracy and an AUC of 0.9293. Abu Adla et al. [9] used a hybrid method of filtering, that is, ANOVA filtering and Sequential Forward Floating Selection (SFFS). Their linear SVM had an accuracy and precision of 91.6 and 93.66 respectively but they did not report F1-scores and the area under the ROC hence it cannot be compared directly. Nested CV and XAI methods were also not used. Batra and Nelson [10] employed correlation-based feature selection where 7-fold cross-validation and final hold-out set were used. The accuracy of RF was the highest with 92.02 and an AUC of 0.947 and the accuracy and AUC of LR was 90.18 and 0.945, respectively, with SVM being the lowest performing. No mechanisms of interpretability were implemented. Thakre et al. [11] presented PCOcare, in which chi-square selection was used to shrink to 30 features among 41. RF outperformed all other models with a ~90.9 accuracy and ~0.89 AUC in both 5-fold CV and test set, which shows that the model has high recall of PCOS-positive samples. SVM and LR did decently and Gaussian Naive Bayes was the weakest. Another study, by Munjal et al. [12], utilised an AI method known as Genetic Algorithm (GA) to reduce 40 to 9 features. The Extra Trees (ETC) were the best classifier, scoring 88% accuracy and AUC of 0.96, which was better than the RF and Decision Tree baseline that were used. Altogether, these studies support the diagnostic potential of optimized ML systems to PCOS but reveal further gaps in internal validation and explainability that are needed to translate a model into practice.

III. Materials And Methods

The **Fig. 1** is a demonstration of PCOS prediction methodology. It starts with data preprocessing, which consists of imputation, feature removal, and AIM - PDCF transformations, continues with feature engineering and clustering. The training and testing datasets are then defined; a training (80 percent) and testing (20 percent). ROS is used to maintain the imbalance and the feature selection is accomplished by QuantumGraphRFE in a hybrid manner. RF, ETC, KNN, XGB are used in combination in a Stacking Ensemble (SE) model that is trained using nested cross-validation and hyperparameter optimization. Lastly, the performance metrics are combined with XAI-based interpretability as a way of ensuring clinical transparency through model assessment.

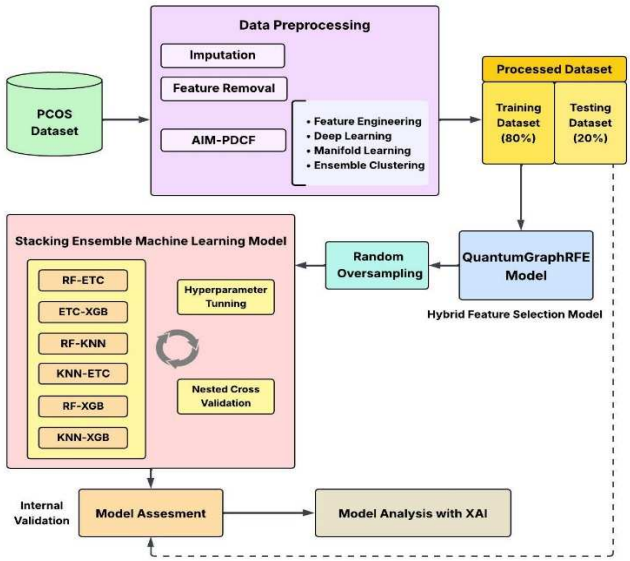


Fig. 1. The Methodological Framework for PCOS Prediction

A. Dataset Description

The following research is based on PCOS Prediction Dataset of Kaggle [13] with the following key properties outlined in **TABLE I**. It has 44 features and the target column called PCOS (Y/N) as a PCOS classification parameter in 541 records of ten Indian hospitals, with demographic, biological, and lifestyle data. The data set shows a severe class imbalance (364 normal; 177 PCOS-positive).

TABLE I. Dataset Characteristics

Characteristic	**Value**
Type	Multivariate
Task	Classification
Instances	541
Features	44
Feature Types	Categorical, Integer, Real

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:55:55 UTC from IEEE Xplore. Restrictions apply.

Categorical Features	17
Continuous Features	27
Target Variable	PCOS Diagnosis (0 = Normal, 1 = PCOS Positive)
Class Distribution	Imbalanced (364 Normal, 177 PCOS Positive)
Data Source	Kaggle Repository (Medical records from 10 hospitals in Kerala, India)

B. Dataset Cleaning and Preprocessing

The variable preprocessing process resulted in high data integrity of 541 patient records and 41 predictive features. Three deductive variables--Sl. No., Patient File No., and an empty column-have been eliminated as being ineffective predictors. The missing values were solved by using simple imputer implemented in scikit-learn, and the method of imputation was the simple mean of numerical attributes and mode at the categorical ones. Type errors (i.e., converting II beta-HCG [mIU/mL] and AMH [ng/mL] to floating-point values to prevent their involvement in computational errors) were also corrected. Preparation of the data was done with Advanced Intelligent Medical Pattern Discovery and Clustering Framework (AIM-PDCF) which is a novel development of this work. AIM-PDCF contributed an Advanced Intelligent Categorical Converter that converted the continuous features into optimal category spaces with distribution-sensitive methods. Forty-one features were treated: 14 processed using optimal_bins, 12 skewed attributes using skewed_tertile, 7 binary, 6 using binary_fallback, and one using medical_quartile and quartile. A 16-dimensional latent space generated by an attention autoencoder was clustered with UMAP-based ensemble clustering into four patient clusters. These had a consensus silhouette value of 0.4654, ARI 0.0511 and NMI 0.0487. Cluster separability, consistent with the notion of clinically relevant PCOS phenotypes, was revealed by clusters as shown in **Fig. 2** (UMAP Embedding). The AIM-PDCF conversion process flow (**Fig. 3**) is a logically outlined process of conversion strategies and fallback mechanisms. Lastly, data was stratified 80/20 partitioned in feature selection, over-sampling randomly, and SE assessments.

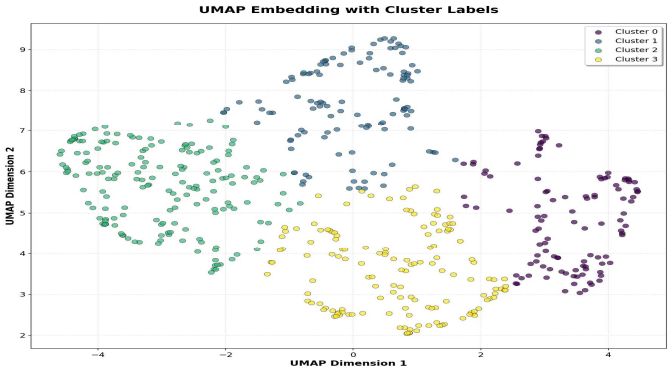


Fig. 2. Cluster Structure in UMAP Latent Space

C. Hybrid Feature Selection Model (QuantumGraphRFE Model)

The Quantum Approximate Quality Assessment (QAQA) with Prim Minimum Spanning Tree and Recursive Feature Elimination Cross- Validation (QuantumGraphRFE) model, an **Algorithm 1**, suggests a new and high-principled amalgamation of diverse computational paradigms. The approach combines quantum-motivated optimization heuristics, graph-theoretical MST construction, and recursive feature elimination in typically achieving very sharp dimensionality reductions by convergence in multiple MSTs with quantum-tunneling perturbations.

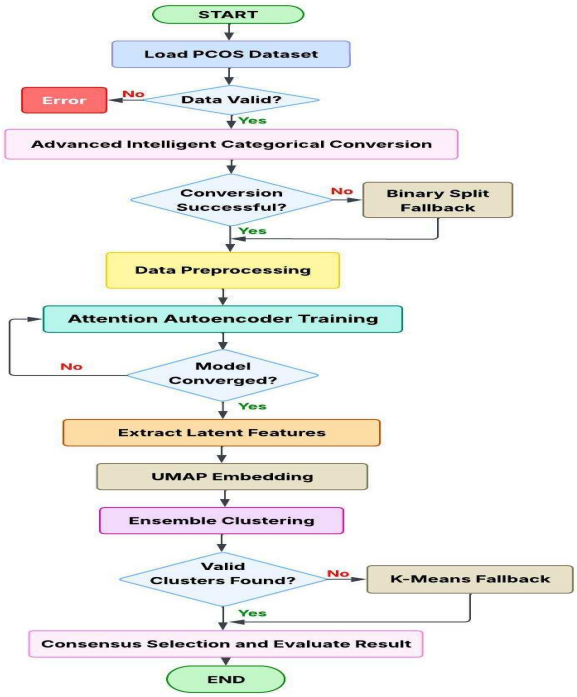


Fig. 3. Workflow for AIM-PDCF's Novel Categorical Converter

Algorithm 1. QuantumGraph-RFE (QAQA-Prim-RFECV) Model

Input: Dataset D = {X, y}, where X ∈ $\mathbb{R}^{nxd}$, y ∈ {0, 1, ..., c-1}
Output: Selected feature indices F_final ⊆ {1, 2, ..., d}
Parameters: α = 0.3, β = 0.4, γ = 0.3, max_iterations = 50
1: Compute QAQA scores: qaqa_scores[i] = 0.5×|corr(X[:,i], y)| + 0.3×interference + 0.2×variability
2: Build MST: mst_structure = minimum_spanning_tree(1 - |corr(X^T)|)
3: Compute MST scores: mst_scores = degree_centrality + edge_weights
4: Train RFECV: rfecv.fit(X, y), rfecv_scores = feature_importances
5: Normalize all scores: qaqa_norm, mst_norm, rfecv_norm
6: Initialize: final_scores = random(d)
7: for iteration = 1 to max_iterations do 8: mst_influence = neighborhood_influence(mst_structure, final_scores)
9: final_scores = α×(mst_norm + mst_influence) + β×rfecv_norm + γ×qaqa_norm
10: if iteration mod 10 = 0 then quantum_tunneling(final_scores)
11: if converged then break
12: end for
13: F_final = top_k_features(final_scores), return F_final

After the entire predictors, it retained 20 salient features (retention rate = 48.8%) at a threshold of 0.281227. Waist (inch) had the highest discriminative capacity (0.712) with Fast food (Y/N) having low discriminative scores (0.013). In the example shown in **Fig. 4**, the Feature Scores Heatmap shows the top 20 features, making it clear how heterogeneous changes

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:55:55 UTC from IEEE Xplore. Restrictions apply.

are represented episodically and novelty and nobility of this multi-paradigmatic approach take center stage.

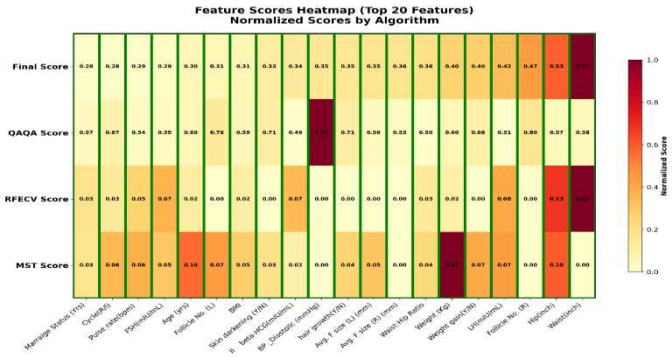


Fig. 4. Heatmap of Feature Rankings - QuantumGraph-RFE Model

D. Class Imbalance Handling

To counter the training set class imbalance (2.06:1; 291 Normal vs 141 PCOS), we used Random Over-Sampling (ROS). This produced 150 new synthesized minority samples attaining a 1:1 ratio (291 per class). In total there were 582 samples to be used in training. The ROS achieved a 51.5 reduction in imbalance, and the model is hence not biased towards the majority class.

E. Stacking Ensemble Machine Learning Models

A hierarchical SE model was used in order to maximize predictive rigor; at Level 1, base learners were stacked under a meta-learner, Logistic Regression (LR). Level 0 consisted of six algorithmic dyads, 'RF–ETC', 'ETC–XGB', 'RF–KNN', 'KNN–ETC', 'RF–XGB' and 'KNN–XGB'. KNN was used together with StandardScaler to normalize the manifolds. Stratified Nested Cross-Validation (k=5 outer, k=3 inner) was applied to coordinate simultaneous performance evaluation and hyperparameter optimization with the help of RandomizedSearchCV. The parametric lattice contained seven major dimensions, with three values (estimators, depth, learning rate, neighbors, C of LR, etc.), so that it yielded 2,187 admissible instantiations (3^7). In the terminal tuning cycle, some 400 or 450 candidate ensembles were traversed. The final evaluation was carried out on hidden test data with instantiated globally optimal hyperparameters.

F. Performance Metrics

The model was evaluated by simple measures from the confusion matrix (Accuracy, Precision, Recall, F1-Score) [14]. In addition, we estimated ROC-AUC to yield global discriminative capacity across all classification thresholds.

IV. Results And Discussion

All of the six SEs reached high diagnostic performance in the test set presented in **TABLE II**. The two best performing models were the 'RF-KNN' and 'KNN-XGB' with the same accuracy of 96.58%. However 'KNN-XGB' performed better with a maximum F1-Score (96.60%), recall (97.26%) and an almost perfect ROC-AUC of 99.59%. On the other hand the 'ETC-XGB' model which had a perfect precision did not have a good recall (84.93%), so was less dependable. Other three models ('RF-ETC', 'KNN-ETC' and 'RF-XGB') were equally high with 93.84% accuracy. Thus, the 'KNN-XGB' model is considered best classifier providing a trade-off between diagnostic performance and stability, as demonstrated by its excellent ROC-AUC.

TABLE II. Performance Evaluation Of Classifiers

Model Name	Accuracy	Precision	Recall	F1 Score	ROC-AUC
RF-ETC	93.84%	97.06%	90.41%	93.62%	97.32%
ETC-XGB	92.47%	100.00%	84.93%	91.85%	98.57%
RF-KNN	96.58%	97.22%	95.89%	96.55%	98.58%
KNN-ETC	93.84%	98.48%	89.04%	93.53%	98.56%
RF-XGB	93.84%	98.48%	89.04%	93.53%	98.40%
KNN-XGB	96.58%	95.95%	97.26%	96.60%	99.59%

Best Performing Model:

KNN-XGB SE model had the best performance overall. Its confusion matrix (**Fig. 5**) showed a remarkable performance with only three False Negatives and two False Positives. This accuracy can also be supported by ROC analysis (**Fig. 6**) and with a close-to-perfect AUC of 0.996 which describes an outstanding discriminative ability. The robustness of the model was validated by a low standard deviation in the 5-fold nested cross-validation (**TABLE III**).

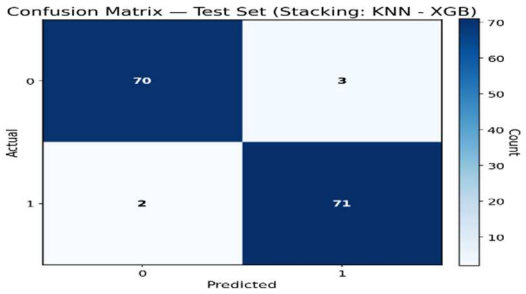


Fig. 5. Confusion Matrix for the Best Model (KNN-XGB)

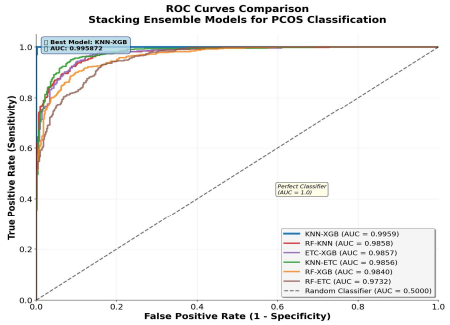


Fig. 6. Comparison of ROC Curves for SE Models

TABLE III. Matrics Across 5 Folds For Knn-xgb Model

	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Fold 1	0.971	0.9577	0.9855	0.9714	0.9977
Fold 2	0.9493	0.9079	1	0.9517	0.9813
Fold 3	0.9203	0.9028	0.942	0.922	0.9771
Fold 4	0.9275	0.8933	0.971	0.9306	0.992

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:55:55 UTC from IEEE Xplore. Restrictions apply.

Fold 5	0.8768	0.9048	0.8382	0.8702	0.9523
Mean	0.929	0.9133	0.9474	0.9292	0.9801
Standard Deviation	0.0353	0.0254	0.0647	0.0382	0.0176
95% Confidence Interval	±0.0309	±0.0222	±0.0567	±0.0335	±0.0154

The strength of the model comes from the interaction between the base learners which have the optimized hyperparameters listed in **TABLE IV**. KNN is good at local, instance-level similarities and XGBoost is good for comprehensive and complex global patterns. Subsequently, the LR meta-learner combines the probabilistic predictions obtained from these multimodal learners to build an ultimate strong classifier with improved generalization and prediction performance.

TABLE IV. FINAL HYPERPARAMETER CONFIGURATION OF THE BEST PERFORMING MODEL (KNN-XGB)

Model	**Hyperparameter**	**Best Value**
KNN	n_neighbors	7
	weights	distance'
	metric	manhattan'
XGBoost	n_estimators	300
	max_depth	9
	learning_rate	0.05
	subsample	1
Logistic Regression (Meta-Learner)	C (Inverse Regularization)	1

Performance Analysis with Explainable AI:

For the sake of transparency and trust in the diagnostic system, the best performing model (KNN-XGB) was interpreted by different XAI methods. Global feature importance analyses, including an averaged LIME score (**Fig. 7**), obtained the three most important predictors for PCOS diagnosis are Follicle_No_R, Cycle_R_I, and Weight_gain_Y_N. Local explanations validate that these traits are equally used to make each human prediction. The SHAP waterfall chart in **Fig. 8**, which presents the effect of the feature values on the output of the model for a single patient. For instance, a positive high value of Follicle_No_R strongly tend to a PCOS diagnosis (+0.202 log-odds). Similarly, the LIME analysis (**Fig. 9**) bring out the contribution of Follicle_No_R in the final prediction.The SHAP heatmap in **Fig. 10** also indicates the trend among various patients, where the contribution of Follicle_No_R remains consistently beneficial in predicting PCOS. The model's reliance on clinical features enhances trust. Cycle irregularity and increased follicle count serve as the Rotterdam criteria for core PCOS features [15-16]. Weight gain, while not diagnostic criteria, is a common symptom linked to insulin resistance and hormonal imbalances in PCOS [17-18]. The model also presents elevated LH levels as a crucial biomarker [19-20]. This correlation with clinical features justifies the use of the model as a diagnostic aid.

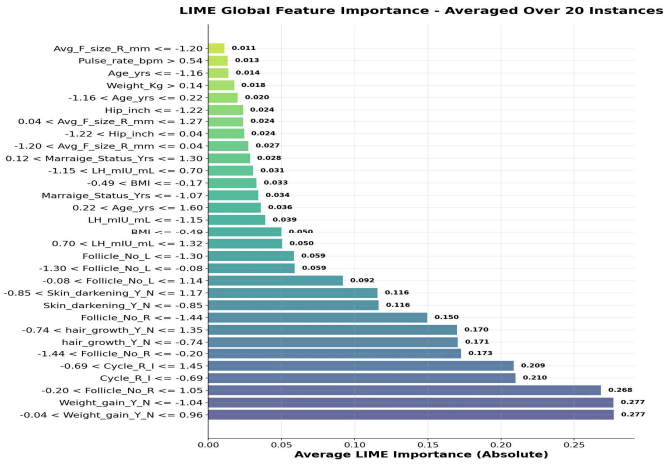


Fig. 7: LIME Global Feature Importance

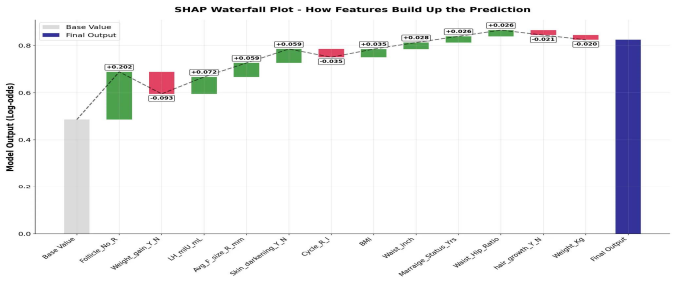


Fig. 8: Local Feature Attribution via SHAP Waterfall Plot

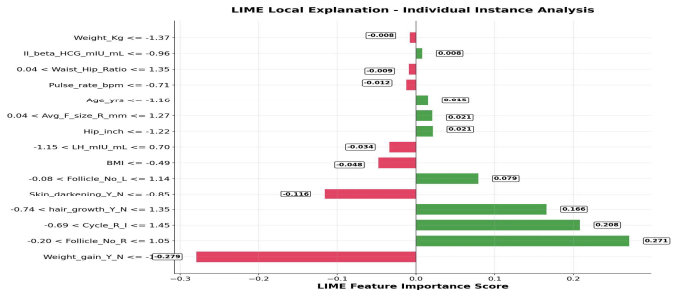


Fig. 9: Local Feature Attribution for a Sample Prediction via LIME

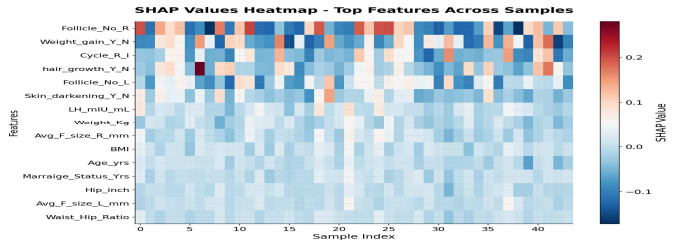


Fig. 10: SHAP Value Heatmap Across Multiple Patient Samples

Comparison with Other Related Works :

We provide a thorough empirical benchmark in **TABLE V**, where our proposed SE with KNN-XGB model (96.58% accuracy, 99.59% ROC-AUC) achieves a superior performance over the existing models having used the same PCOS dataset. It is higher than the previously best reported accuracy of 94% by

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:55:55 UTC from IEEE Xplore. Restrictions apply.

Rahman et al. Our model is the only one to incorporate a novel, hybrid feature selection method, the QuantumGraph-RFE Model which ensures robustness through nested cross-validation. Most importantly, it is the only method that utilises XAI techniques for model interpretability, which is a key missing factor in prior research.

TABLE V. PROPOSED MODEL COMPARISON WITH RELETED WORKS

Existing Literatures	Dataset of	Classification Method	Accuracy	ROC-AUC	Feature Selection Model	Nested Cross Validation	XAI
Sreejith et al. [7]	541 Patients	RF-RDA	89.81%	-	✕	✕	✕
Rahman et al. [8]	541 Patients	AdaBoost	94%	92.93%	MI Model	✕	✕
Abu Adla et al. [9]	541 Patients	Linear SVM	91.60%	-	✕	✕	✕
Batra and Nelson [10]	541 Patients	RF	92.02%	94.70%	✕	✕	✕
Thakre et al. [11]	541 Patients	RF	90.91%	89.00%	✕	✕	✕
Munjal et al. [12]	541 Patients	ETC	88.00%	96.00%	✕	✕	✕
Proposed Model	**541 Patients**	**KNN+XGB**	**96.58%**	**99.59%**	**QuantumGraph-RFE Model**	✓	✓

V. CONCLUSION

In this research, we have created and tested a new stacking ensemble ML platform to diagnose PCOS. We combine a novel AIM-PDCF model of the intelligent data preprocessing and a QuantumGraphRFE data model of the multi-paradigm feature selection which is highly effective. The proposed KNN-XGB SE model with accuracy of 96.58% and ROC-AUC of 99.59% is superior to state-of-the-art in this dataset. Notably, the framework focuses on clinical AI transparency through the use of Explainable AI (XAI). The interpretability of the model with the help of the LIME and SHAP analyses confirmed that the decisions of the model were based on the clinically relevant indicators of follicle numbers, cycle abnormalities, and weight gain, which added trust and encouraged clinical adoption. Intense nested cross-validation protocol also provides stability and generalization of the models used. This study offers a promising instrument to support the early diagnosis of PCOS by being accurate and interpretable. Subsequent efforts will be directed to validation using larger and multi-centric datasets and possible use in real-time clinical decision support systems.

REFERENCES

[1] World Health Organization, "Polycystic ovary syndrome," Fact Sheet, Feb. 7, 2025. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/polycystic-ovary-syndrome

[2] A. A. Farhana and J. Shi, "A narrative review on global epidemiology of PCOS and its hormonal management," SSR Inst. Int. J. Life Sci., vol. 10, no. 3, pp. 5516–5521, 2024, doi: 10.21276/SSR-IIJLS.2024.10.2.22

[3] World Heart Federation, "Women & CVD." [Online]. Available: https://world-heart-federation.org/what-we-do/women-cvd/

[4] S. Nasim, M. S. Almutairi, K. Munir, A. Raza, and F. Younas, "A novel approach for polycystic ovary syndrome prediction using machine learning in bioinformatics," IEEE Access, vol. 10, pp. 97610–97624, 2022. doi: 10.1109/ACCESS.2022.3205587

[5] M. Y. Lu, W. L. Chuang, and M. L. Yu, "The role of artificial intelligence in the management of liver diseases," Kaohsiung J. Med. Sci., vol. 40, no. 11, pp. 962–971, 2024, doi: 10.1002/kjm2.12513.

[6] V. V. Khanna, K. Chadaga, N. Sampathila, S. Prabhu, V. Bhandage, and G. K. Hegde, "A distinctive explainable machine learning framework for detection of polycystic ovary syndrome," Appl. Syst. Innov., vol. 6, no. 2, Art. no. 32, 2023, doi: 10.3390/asi6020032.

[7] S. Sreejith, H. K. Nehemiah, and A. Kannan, "A clinical decision support system for polycystic ovarian syndrome using red deer algorithm and random forest classifier," Healthcare Analytics, vol. 2, p. 100102, 2022, doi: 10.1016/j.health.2022.100102

[8] M. M. Rahman, A. Islam, F. Islam, M. Zaman, M. R. Islam, M. A. Sakib, and H. H. Babu, "Empowering early detection: A web-based machine learning approach for PCOS prediction," Informatics in Medicine Unlocked, vol. 47, p. 101500, 2024, doi: 10.1016/j.imu.2024.101500

[9] Y. A. A. Adla, D. G. Raydan, M. Z. J. Charaf, R. A. Saad, J. Nasreddine, and M. O. Diab, "Automated detection of polycystic ovary syndrome using machine learning techniques," in 2021 Sixth International Conference on Advances in Biomedical Engineering (ICABME), Oct. 2021, pp. 208–212, doi: 10.1109/ICABME53305.2021.9604506

[10] H. Batra and L. Nelson, "DCADS: data-driven computer aided diagnostic system using machine learning techniques for polycystic ovary syndrome," International Journal of Performability Engineering, vol. 19, no. 3, p. 193, Mar. 2023, doi: 10.23940/ijpe.23.03.p5.193202

[11] V. Thakre, S. Vedpathak, K. Thakre, and S. Sonawani, "PCOcare: PCOS detection and prediction using machine learning algorithms," Bioscience Biotechnology Research Communications, vol. 13, no. 14, pp. 240–244, 2020, doi: 10.21786/bbrc/13.14/56

[12] A. Munjal, R. Khandia, and B. Gautam, "A machine learning approach for selection of polycystic ovarian syndrome (PCOS) attributes and comparing different classifier performance with the help of WEKA and PyCaret," International Journal of Scientific Research, vol. 9, no. 12, pp. 59–63, Dec. 2020, doi: 10.36106/ijsr/20071512

[13] P. Kottarathil, "Polycystic ovary syndrome (PCOS)," Kaggle, 2020. [Online].Available:https://www.kaggle.com/datasets/prasoonkottarathil/polycystic-ovary-syndrome-pcos. [Accessed: Jan. 10, 2025].

[14] M. Sokolova and G. Lapalme, "A systematic analysis of performance measures for classification tasks," Information Processing and Management, vol. 45, no. 4, pp. 427-437, 2009

[15] S. M. Sirmans and K. A. Pate, "Epidemiology, diagnosis, and management of polycystic ovary syndrome," Clin. Epidemiol., vol. 6, pp. 1–13, Dec. 2013.

[16] H. S. Kim and J. Y. Rhee, "Update on polycystic ovary syndrome," Chonnam Med. J., vol. 55, no. 1, pp. 1–7, 2019.

[17] A. J. Dapas and E. F. Schisterman, "Obesity and polycystic ovary syndrome: Implications for pathogenesis and novel management strategies," Curr. Obes. Rep., vol. 8, no. 4, pp. 419–429, Sep. 2019.

[18] M. Escobar-Morreale, "Polycystic ovary syndrome: definition, diagnosis, and treatment," Nature Reviews Endocrinology, vol. 14, no. 5, pp. 270–284, May 2018. doi: 10.1038/nrendo.2018.24.

[19] R. Pastor-Anglada and M. A. Martínez, "Hypothalamic–pituitary–ovarian dysfunction in polycystic ovary syndrome: the role of GnRH pulse frequency," Front. Endocrinol., vol. 12, art. 667422, Jun. 2021.

[20] "Mechanism of elevated LH/FSH ratio in lean PCOS revisited: a neuroendocrine path analysis," Sci. Rep., vol. 14, art. 8229, 2024. [Online]. Available: https://www.nature.com/articles/s41598-024-58064-0. [Accessed: Aug. 27, 2025].

Authorized licensed use limited to: Southeast University. Downloaded on October 18,2026 at 04:55:55 UTC from IEEE Xplore. Restrictions apply.